



Probability and Random Variables

What You Should Be Able to Do

- Explain why probability is needed in stochastic simulation.
- Distinguish outcomes, events, and random variables.
- Compute and interpret mean, variance, and standard deviation.
- Choose a plausible distribution and test it with NumPy.

Why Probability Matters in Simulation

Meaning: Probability describes uncertainty using possible outcomes and their likelihoods.

In a model: Random inputs make repeated runs produce different outputs.

Example: Customers do not arrive at perfectly regular times, and service durations vary.

Takeaway: Probability lets the model reproduce variability instead of hiding it inside one average.

Where Does Randomness Come From?

- **Demand:** the number of orders, calls, patients, or packets changes over time.
- **Duration:** service, repair, travel, and processing times are not identical.
- **Outcome:** a component may pass, fail, recover, or require rework.
- **Human behavior:** people may choose different actions in the same situation.

Modelling question: Which sources of variation can change the decision?

Four Probability Words to Know

- **Experiment:** a repeatable process whose result is uncertain, such as rolling a die.
- **Outcome:** one possible result, such as rolling a 4.
- **Sample space:** the set of all possible outcomes, such as $\{1, 2, 3, 4, 5, 6\}$.
- **Event:** one or more outcomes of interest, such as rolling an even number.

Worked Example: One Fair Die

Experiment: Roll one fair six-sided die.

Sample space: $S = \{1, 2, 3, 4, 5, 6\}$.

Event A: Roll an even number, so $A = \{2, 4, 6\}$.

Calculation: $P(A) = \text{favorable outcomes} / \text{all outcomes} = 3 / 6 = 0.50$.

Interpretation: In many rolls, about half are expected to be even.

Three Basic Probability Rules

- **Range:** Every probability is between 0 and 1.
 - **Total:** The probabilities of all mutually exclusive outcomes sum to 1.
 - **Impossible and certain:** An impossible event has probability 0; a certain event has probability 1.
- Check:** A proposed distribution is invalid if it contains a negative probability or does not total 1.

The Complement Rule

Meaning: A^c means that event A does not occur.

Rule: $P(A^c) = 1 - P(A)$.

Example: If $P(\text{server failure}) = 0.08$, then $P(\text{no failure}) = 1 - 0.08 = 0.92$.

Why useful: It is often easier to calculate the probability of the opposite event.

The Addition Rule

Meaning: A or B includes outcomes in A, in B, or in both.

General rule: $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$.

Reason: The overlap is subtracted because it was counted twice.

Special case: If A and B cannot occur together, $P(A \text{ and } B) = 0$.

Conditional Probability

Meaning: $P(A \mid B)$ asks for the chance of A after learning that B occurred.

Rule: $P(A \mid B) = P(A \text{ and } B) / P(B)$, provided $P(B) > 0$.

Example: Failure risk may be higher given that a server is already overheating.

Simulation use: Conditional distributions allow model behavior to change with the current state.

Independence Is a Strong Assumption

Independent: Knowing B occurred does not change the probability of A.

Test: If independent, $P(A \mid B) = P(A)$ and $P(A \text{ and } B) = P(A)P(B)$.

Example: Two component failures may not be independent if both depend on temperature.

Takeaway: Do not assume independence only because it makes the model easier.

Quick Check: Which Rule Applies?

1. Find the chance that a packet is not lost. Use the complement rule.
2. Find the chance of error A or error B. Use the addition rule.
3. Find failure risk given overheating. Use conditional probability.
4. Decide whether one event changes another. Check independence.

What Is a Random Variable?

Definition: A random variable assigns a number to each outcome of an uncertain experiment.

Example experiment: Observe customer arrivals during the next ten minutes.

Random variable: X = number of arrivals observed.

Possible values: X can be 0, 1, 2, 3, and so on.

Key idea: The variable is numerical; its value is unknown before the experiment.

Discrete or Continuous?

- **Discrete:** values can be listed or counted, such as 0, 1, 2, 3 arrivals.
- **Continuous:** values can fall anywhere in an interval, such as 3.27 minutes.

Question to ask: Am I counting separate items, or measuring on a scale?

Important: A rounded measurement is still conceptually continuous.

PMF: Probabilities for Discrete Values

Name: PMF means probability mass function.

Purpose: It assigns $P(X = x)$ to each possible discrete value x .

Requirements: Every probability is nonnegative and all probabilities sum to 1.

Example: If X is the number of defects, the PMF may assign probabilities to 0, 1, 2, and 3 defects.

PDF: Density for Continuous Values

Name: PDF means probability density function.

Purpose: Probability over an interval equals the area under the density curve.

Important: For a continuous variable, $P(X = \text{exactly } x) = 0$.

Example: We calculate $P(3 < \text{service time} < 5)$, not the chance of exactly 4.000000 minutes.

CDF: Probability Up to a Value

Name: CDF means cumulative distribution function.

Rule: $F(x) = P(X \leq x)$.

Meaning: It accumulates probability from the smallest possible value up to x .

Example: $F(5) = 0.80$ means 80% of values are at or below 5.

Use: Percentiles and random-variate generation rely on cumulative probability.

Expected Value Describes the Center

Meaning: The expected value is the long-run probability-weighted average.

Discrete rule: $E[X] = \sum xP(X = x)$.

Interpretation: Across many repetitions, the sample average should approach $E[X]$.

Caution: The expected value need not be a possible single outcome.

Variance and Standard Deviation

Variance: $\text{Var}(X) = E[(X - \mu)^2]$, the average squared distance from the mean.

Shortcut: $\text{Var}(X) = E[X^2] - \mu^2$.

Standard deviation: $\sigma = \sqrt{\text{Var}(X)}$, which returns to the original unit.

Why it matters: Two inputs with the same mean can create very different congestion or risk.

Worked Distribution: Daily Demand

Random variable: D = daily demand for a product.

Possible values: $D = 1, 2, 3, \text{ or } 4$ units.

Probabilities: $P = 1/6, 1/3, 1/3, \text{ and } 1/6$, respectively.

Validity check: $1/6 + 1/3 + 1/3 + 1/6 = 1$.

Goal: Compute the mean and standard deviation, then interpret them.

Step 1: Calculate Mean Demand

Rule: $E[D] = \sum dP(D = d)$.

Substitute: $E[D] = 1(1/6) + 2(1/3) + 3(1/3) + 4(1/6)$.

Add: $E[D] = 1/6 + 2/3 + 1 + 2/3 = 2.50$.

Interpretation: Average demand approaches 2.5 units per day over many days.

Step 2: Calculate Demand Variance

First: $E[D^2] = 1^2(1/6) + 2^2(1/3) + 3^2(1/3) + 4^2(1/6)$.

Result: $E[D^2] = 43/6 \approx 7.1667$.

Variance: $\text{Var}(D) = E[D^2] - E[D]^2 = 7.1667 - 2.5^2 \approx 0.9167$.

Standard deviation: $\sigma = \sqrt{0.9167} \approx 0.9574$ units.

Step 3: Interpret the Result

Center: The long-run average is 2.5 units per day.

Spread: A typical distance from the mean is about 0.96 unit.

Not a prediction: Tomorrow's demand is still one of 1, 2, 3, or 4 units.

Simulation use: The full distribution generates daily demand; the mean and standard deviation summarize it.

Uniform Distribution

Meaning: All equal-length intervals between minimum a and maximum b have equal probability.

Parameters: The distribution is defined by its lower and upper bounds.

Example: A start time is equally likely anywhere from 8:00 to 8:10.

Use with care: Uniform is reasonable only when there is evidence that no region is more likely.

Exponential Distribution

Meaning: A positive, right-skewed distribution often used for time between events.

Parameter: Its scale is the mean waiting or interarrival time.

Example: Time between independent arrivals when the average rate is constant.

Shape: Short times are common, while long times are possible but less frequent.

The Memoryless Property

Meaning: For an exponential time, time already waited does not change the remaining-time distribution.

Example: If arrivals truly follow the assumptions, waiting five minutes does not make an arrival due.

Why strong: Many real processes age, tire, warm up, or become more likely to finish over time.

Takeaway: Use exponential only when its process assumptions are defensible.

Normal Distribution

Meaning: A symmetric, bell-shaped distribution centered at mean μ .

Parameters: Mean μ controls the center; standard deviation σ controls spread.

Example: Measurement error created by many small independent influences.

Caution: Normal values are unbounded, so it may generate impossible negative times.

Match the Distribution to the Process

- **Bernoulli:** one trial with two outcomes, such as pass or fail.
- **Binomial:** number of successes in a fixed number of independent trials.
- **Poisson:** count of events in an interval under a constant average rate.
- **Exponential:** positive time between independent events under a constant rate.
- **Normal:** symmetric measurement variation around a mean.

Read a Histogram as Evidence

- **Center:** where most observations are located.
- **Spread:** how widely observations vary.
- **Shape:** symmetric, right-skewed, left-skewed, or multimodal.
- **Unusual values:** gaps or outliers that may indicate errors or rare behavior.

Caution: Histogram appearance changes with bin width and sample size.

Why the Mean Alone Can Mislead

Problem: Replacing a random time with its mean removes variability.

Queue effect: Occasional clusters of arrivals can create waits even when the average looks manageable.

Risk effect: Rare large values can dominate cost, delay, or system failure.

Takeaway: Model the distribution when variation affects the performance measure or decision.

Same Mean, Different Queue Delays

Weibull service times: average queue delay ≈ 4.36 minutes.

Gamma service times: average queue delay ≈ 4.54 minutes.

Normal service times: average queue delay ≈ 6.04 minutes.

Exponential service times: average queue delay ≈ 6.71 minutes.

Lognormal service times: average queue delay ≈ 7.19 minutes.

A Distribution Selection Checklist

1. Define the variable, unit, and process that generates it.
2. Check support: count or measurement, bounded or unbounded, positive or negative.
3. Inspect data for center, spread, shape, outliers, and time patterns.
4. Compare candidate assumptions and goodness of fit.
5. Test whether the simulation decision changes under alternatives.

Dependence and Correlation Matter

Dependence: Knowing one value changes what is likely for another value.

Positive correlation: high values tend to occur with high values.

Negative correlation: high values tend to occur with low values.

No linear correlation: does not guarantee independence.

Modelling risk: Sampling dependent inputs independently can create unrealistic combinations.

Queue Outputs Are Often Correlated

Example: A long queue now often implies a long queue a moment later.

Reason: The system state carries information from one observation to the next.

Consequence: Ten nearby observations do not contain as much independent information as ten separate replications.

Practice: Use appropriate warm-up, run length, replications, and output analysis.

Correlation Changes Statistical Precision

If ignored: Standard errors may be too small and confidence intervals too narrow.

Why: Correlated observations contribute less new information than independent observations.

Safe habit: Separate independent replications and preserve the random seed or stream design.

Preview: Later output analysis will estimate uncertainty using valid simulation evidence.

Lab Exercise: Simulate Daily Demand

1. Define $D = \{1, 2, 3, 4\}$ and probabilities $\{1/6, 1/3, 1/3, 1/6\}$.
 2. Generate 10,000 observations using a documented NumPy seed.
 3. Compute simulated mean and population standard deviation.
 4. Compare the simulation with $\mu = 2.5$ and $\sigma \approx 0.9574$.
- Purpose:** Show that repeated sampling reproduces the theoretical distribution.

Lab Code: Generate the Sample

```
import numpy as np
values = np.array([1, 2, 3, 4])
probabilities = np.array([1/6, 1/3, 1/3, 1/6])
rng = np.random.default_rng(142)
samples = rng.choice(
    values, size=10000, p=probabilities
)
```

Lab Code: Compare the Statistics

```
theoretical_mean = 2.5
theoretical_sd = np.sqrt(11/12)
simulated_mean = samples.mean()
simulated_sd = samples.std(ddof=0)
print(theoretical_mean, theoretical_sd)
print(simulated_mean, simulated_sd)
# Seed 142: 2.5074 and 0.95684 approximately
```

Lab Result: How to Explain It

Theoretical: Mean = 2.5000; standard deviation $\approx$ 0.9574.

Simulated: With seed 142, mean = 2.5074; standard deviation $\approx$ 0.9568.

Difference: The values are close but not identical because the sample is random and finite.

Conclusion: The generated sample is consistent with the intended demand distribution.

Do not write: The code is correct only because the means match exactly; exact equality is not expected.

Week 3 Lab Assessment Checklist

- Submit one runnable notebook containing the variable, values, probabilities, and seed.
- Generate 10,000 observations and show the code output.
- Report theoretical and simulated mean and standard deviation.
- Write a short interpretation of agreement, variation, and limitations.
- Keep code, output, and explanation together; be ready to explain every line.

Week 3 Understanding Checklist

- I can distinguish an outcome, event, and random variable.
- I can classify a random variable as discrete or continuous.
- I can read a PMF, PDF, and CDF at an introductory level.
- I can calculate and interpret mean, variance, and standard deviation.
- I can justify a distribution using the process, data, and assumptions.

Next Week: Random Numbers and Monte Carlo

Question: How does a computer generate repeatable pseudo-random numbers?

Method: Convert uniform random numbers into samples from chosen distributions.

Application: Repeat uncertain experiments to estimate probabilities, risks, and expected outcomes.

Bring: Your completed Week 3 notebook and its documented seed.

Bridge: This week defined the distributions; next week generates and uses the samples.

Thank you!